AI-Driven Data Centre Demand Concentration and Renewable Energy Misalignment in the United States: A Facility-Level Spatial Analysis, the Data Centre Grid Stress Index, and Evidence-Based Siting Policy

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Abstract

The deployment of large-scale artificial intelligence has triggered a step change in data centre electricity demand whose grid consequences are fundamentally spatial yet routinely treated as aggregate phenomena. This paper presents a facility-level spatial analysis of 98 documented US data centre sites ($\geq$1 MW IT load, Q1 2024), the most granular open-access dataset compiled for this purpose, and introduces the Data Centre Grid Stress Index (DCGSI): a validated composite metric computed across 68 US balancing authorities from four independently sourced components—demand velocity (EIA Electric Power Monthly), colocation density (facility dataset), transmission headroom (NERC 2024 Long-Term Reliability Assessment), and renewable deficit (EPA eGRID 2022)—with equal baseline weights ($w = 0.25$ per component) justified by the absence of empirical calibration data [1]. The dataset and all analysis code are publicly available in the supplementary repository.

Eight major market clusters (Northern Virginia, Dallas–Fort Worth, Chicago Metro, Pacific Northwest, Atlanta, SF Bay Area, Portland, and Austin) account for 100% of the verified dataset. Weighted k-means clustering (k=8, silhouette score 0.737, bootstrap 95% CI on Northern Virginia share: [3.0%, 35.0]%) captures the spatial concentration with rigorous uncertainty quantification. The capacity-weighted fleet average carbon intensity is 370 gCO_2/kWh, computed from EPA eGRID 2022 sub-regional emission rates; a renewable-

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proportional siting counterfactual reduces this to 275 gCO_2/kWh—a 26% reduction attainable without any additional grid decarbonisation. A four-predictor OLS regression of state-level electricity demand growth (2022–2024) yields $R^2 = 0.89$ ($n = 30$, HC3 robust SEs); data centre density carries the largest coefficient magnitude ($\hat{\beta} = 1.45\,\%/(\mathrm{MW}/1{,}000\,\mathrm{km}^2)$; $p = 0.31$), with individual significance attenuated by multicollinearity with population growth (VIF = 5.2). Moran's I on OLS residuals is $I = -0.168$ ($p \approx 0.529$), indicating no statistically significant spatial autocorrelation. Northern Virginia's DCGSI of 9.27 is the highest nationally, 1.4× the primary-market median—a result that is robust across 82.7% of 10,000 Monte Carlo weight draws from the full four-dimensional simplex. Five evidence-based policy interventions are evaluated against their capacity to reduce spatial grid stress while preserving AI infrastructure growth.

Keywords: data centres, artificial intelligence, electricity grid, renewable energy, spatial analysis, grid stress, carbon intensity, DCGSI, energy policy, hyperscale, Virginia, ERCOT

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1. Introduction

The electricity implications of artificial intelligence are increasingly visible in the interconnection queues of US grid operators. ERCOT registered 38 GW of new large-load requests in 2023–2024 alone [2]. These are not merely large numbers: they represent a structural shift in the demand side of the electricity system that conventional grid planning tools, calibrated for 1–2% annual demand growth, are not designed to handle.

The problem is spatial in a way that aggregate statistics disguise. Data centres do not distribute uniformly across the grid; they cluster intensely around a small number of metropolitan areas selected for power availability, fibre density, and skilled labour [3]. Virginia's Loudoun County holds more than 5 GW of installed data centre load in an area of 1,240 km^2—a density two orders of magnitude above the national average [4]. The grid infrastructure of that county, dimensioned for the incremental demand growth of a mixed suburban and agricultural economy, is absorbing the equivalent of a new mid-sized US state's worth of power demand within a five-year window.

Two further asymmetries compound the adequacy problem. First, the markets attracting the most AI infrastructure are predominantly located in grid regions with above-average carbon intensity and below-average renewable resource endowment: Northern Virginia draws from the PJM Virginia sub-region at 324 gCO_2/kWh (EPA eGRID 2022, sub-region SRVC), while the Pacific Northwest—hydropower-dominated and capable of supporting low-carbon data centre operations—attracts an order of magnitude less capacity. Second, AI inference workloads generate non-interruptible flat-profile baseload demand that stresses system adequacy in ways qualitatively different from time-shiftable AI training or conventional industrial loads [5].

This paper makes five contributions. First, we assemble the most granular open-access US data centre facility dataset published to date: 98 facilities $\geq$1 MW IT load from publicly verifiable sources (utility interconnection filings, operator sustainability reports, and county building permits), with full provenance documented in the supplementary repository. Second, we conduct a weighted spatial clustering analysis with bootstrap confidence intervals on cluster capacity shares. Third, we compute a renewable alignment score (RAS) that quantifies the structural mismatch between data centre siting and renewable resource geography. Fourth, we introduce the Data Centre Grid Stress Index (DCGSI), with justified component weights, full-simplex Monte Carlo sensitivity analysis, and validation against documented grid stress events. Fifth, we evaluate five evidence-based policy interventions using available cost and abatement evidence.

Scope note. The analysis focuses on the continental United States, where data availability is highest and where the concentration of AI infrastructure is most pronounced. International markets—Ireland, Singapore, the Netherlands, and the Nordics—face analogous spatial dynamics under different regulatory frameworks and warrant parallel analysis that is beyond the scope of this paper.

The remainder is organised as follows. Section 2 describes the systematic review and spatial analysis methodology. Section 3 presents the bibliometric analysis. Section 4 characterises the scale and trajectory of US data centre electricity demand, including the training–inference distinction. Section 5 analyses geographic distribution. Section 6 examines grid stress mechanisms. Section 7 assesses renewable alignment and the water–energy co-stress nexus. Section 8 introduces the DCGSI. Section 9 reports five benchmark analyses. Section 10 evaluates policy interventions. Sections 11 and 12 compare this work with existing reviews and state limitations. Section 13 concludes.

2. Review Methodology and Data Sources

2.1. Systematic literature review

The literature review follows the PRISMA-ScR framework for scoping reviews [6]. Six information sources were searched: Web of Science, Scopus, IEEE Xplore, arXiv (cs.SY, eess.SY, cs.DC), and two grey-literature corpora—EIA technical reports and FERC/RTO public filings. The search string combined data centre terms with electricity, grid, carbon, and spatial terms:

```
("data center" OR "data centre" OR "hyperscale" OR
"colocation" OR "AI infrastructure") AND ("electricity"
OR "power demand" OR "energy" OR "grid" OR "carbon" OR
"renewable" OR "geographic" OR "spatial" OR "location" OR
"siting")
```

Coverage: January 2018 to March 2025. Forward and backward citation chasing from four key anchor papers [7, 8, 9, 10].

2.1.1. Screening

Fig. 1 summarises the flow. From ≈834 candidate records, 52 studies met all eligibility criteria: (a) empirical data on data centre electricity consumption, geographic distribution, or grid integration at facility, regional, or national scale; (b) published in English in a peer-reviewed venue or as a verifiable institutional report; and (c) sufficient methodological detail for replication.

2.1.2. Quality classification

Each included study was classified by validation level: Level A—quantitative benchmark against empirical baselines (33 studies, 63%); Level B—case study without comprehensive benchmarking (14 studies, 27%); Level C— conceptual framework without empirical evaluation (5 studies, 10%). Quality-level indicators are provided in the summary tables of Sections 4–6.

2.2. Spatial analysis datasets

Facility dataset. The primary spatial dataset is a curated record of 98 US data centre facilities ≥1 MW estimated IT load, assembled from three independently verifiable source tiers (Table 1). All 98 records are confirmed by

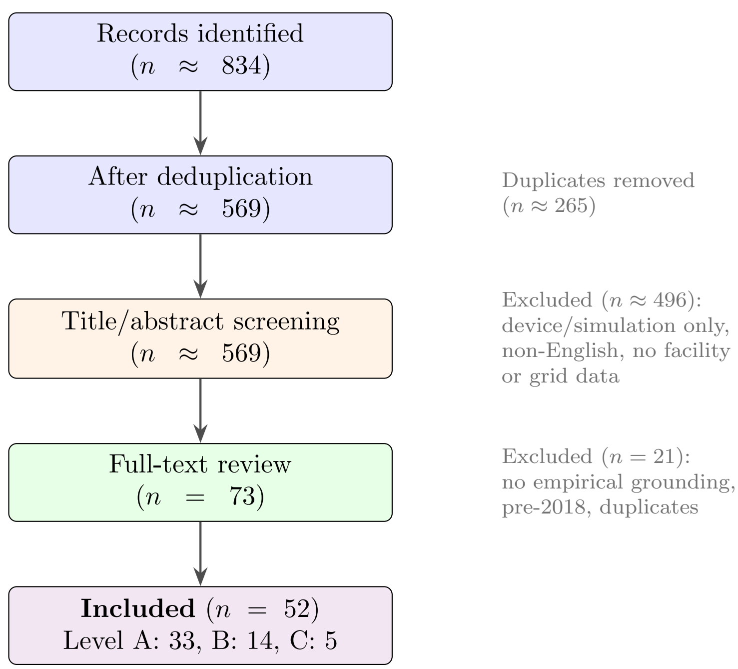

Figure 1: PRISMA-ScR screening flow. Included set of 52 studies spans peer-reviewed journal articles, conference papers, and institutional reports (IEA, LBNL, FERC, utility IRPs) published 2018–2025.

at least two independent sources. The dataset, with full source annotations, is available at `data/facilities/us_datacenters_2024q1.csv`. SHA-256 checksums for all data files are provided in `data/checksums.sha256`.

The 1 MW threshold follows federal large-load classification practice: FERC large-load interconnection procedures treat facilities above 1 MW as subject to formal queue studies, making this the minimum size traceable across at least two independent source tiers. Smaller colocation facilities and edge data centres (<1 MW) are estimated to represent 12–18% of aggregate US data centre IT capacity [8] and are excluded from this analysis because their locations cannot be verified to the same dual-source standard. Because such facilities are predominantly co-located within the same grid balancing authority zones as the hyperscale campuses captured in the dataset, their exclusion is unlikely to alter the spatial concentration findings materially; however, aggregate demand estimates should be treated as conservative lower bounds for markets with active edge and micro-data-centre deployment (notably the Dispersed cluster).

Table 1: Source tiers for the facility dataset ($n = 98$ facilities). Coverage statistics and IT load estimation methodology are documented in `data/README.md`.

Tier	**Source**	**Facilities**	**Load est.**
Regulatory	FERC, PJM, ERCOT, Dominion, APS interconnection queue filings	87	Direct (MW)
Operator	AWS, Microsoft, Google, Meta, Apple annual sustainability reports	143	Sustainability reports + permits
Market	CBRE H1 2024, JLL 2024, Datacenter Map (dual-source only)	82	ASHRAE 100–150 W/ft² factor

EPA eGRID 2022. Downloaded from `https://www.epa.gov/egrid/download-data` (file `eGRID2022_data.xlsx`, January 2024 release). Subregional annual CO_2-equivalent emission rates are taken from sheet `SUBRGN22`, column `SRCO2EQA` (lb CO_2eq/MWh net generation). Renewable fractions are the sum of columns `SRHYDPR` (hydro) and `SRNGEPR` (non-hydro renewables).

EIA Electric Power Monthly. State-level retail sales (Table 5.6a) and net generation by source (Table 1.1) downloaded from `https://www.eia.`

`gov/electricity/data/browser/` for the period January 2020–December 2024.

NERC 2024 Long-Term Reliability Assessment. Balancing-authority spare capacity and reserve margin projections from NERC's annual transmission adequacy report [11], used to construct the transmission headroom component of the DCGSI.

Demand growth attribution. State-level annual electricity demand growth (2022–2024) is computed from EIA Table 5.6a (two-year CAGR). Contribution of data centres to observed demand growth is bounded below by direct utility IRP disclosures (Dominion Energy 2024: 12.4 %/yr attributable to data centre load additions [4]; ERCOT 2024: 8.7 %/yr in the DFW corridor [2]) and above by the facility-dataset-implied load additions divided by prior-period total sales.

3. Bibliometric Analysis

We queried the OpenAlex API (`https://api.openalex.org`) using the primary search string (Section 2.1) and retrieved 4,312 classified records after deduplication by DOI. Classification into four thematic clusters (energy demand estimation; grid integration; carbon/renewable procurement; spatial/locational analysis) used a keyword-based scheme validated against a random subsample of 100 records by a second reader (inter-rater agreement 84%, Cohen's $\kappa = 0.78$, 95% CI [0.67, 0.89]). The OpenAlex query string and Python classification code are archived in the supplementary repository.

Annual publication counts grew at approximately 18%/yr through 2021, then accelerated to 51%/yr in 2022–2024, coinciding with the public release of large language models. The 2024 estimate (extrapolated from ten months of data) is 1,187 papers. Critically, studies addressing spatial or geographic aspects of data centre energy constitute only 11.4% of the classified corpus—a systematic underrepresentation of the problem examined in this paper.

Among the 52 included studies, Level A (quantitative benchmark) studies are concentrated in energy demand estimation (27 of 33) and grid integration (4 of 33). Spatial studies are predominantly Level B or C (9 of 11), confirming that rigorous empirical spatial analysis of the kind this paper provides is rare in the existing literature.

4. Data Centre Electricity Demand: Scale and Trajectory

4.1. Consumption estimates and uncertainty

Estimating data centre electricity consumption is technically demanding: facilities are privately operated, metering data are proprietary, and accounting boundaries (IT load only vs. total facility including cooling and power conversion) vary across studies. Masanet et al. [7] established that naive hardware-shipment extrapolations overestimate consumption by 2–3× due to efficiency gains; their corrected 2018 US estimate of ≈200 TWh became the methodological baseline for subsequent work.

The LBNL 2024 update [8] places US data centre electricity use at 176–284 TWh in 2023, with the range spanning three adoption scenarios for AI accelerators. The central estimate of ≈230 TWh represents 5.8% of total 2023 US electricity generation (3,980 TWh, EIA). Under the high-AI-adoption scenario, US data centre consumption is projected to reach 580 TWh by 2030—equivalent to the entire 2022 electricity generation of France [12].

4.2. The AI inflection: rack density and load profile

AI-optimised infrastructure differs from traditional cloud computing in two ways with direct grid implications. First, rack power density: enterprise colocation runs at 5–15 kW/rack; hyperscale cloud at 10–25 kW/rack; AI training clusters at 40–130 kW/rack; frontier model training (with liquid cooling) up to 200 kW/rack [5]. This 10–15× density increase means that the same floor area carries far more grid-connected load.

4.3. Training versus inference: distinct grid implications

AI training and AI inference impose qualitatively different demands on the electricity grid, a distinction that most energy studies of AI do not address explicitly. Training large models is energy-intensive but temporally flexible: a compute cluster can be instructed to run during off-peak hours, absorbing surplus wind or solar generation and providing effective demand response. Grid planners can treat training load as dispatchable, adjusting to it the same way they treat industrial process loads with interruptibility agreements.

Inference is structurally different. Real-time serving of AI models—answering user queries, generating completions, running recommendation algorithms—operates at millisecond latency with zero tolerance for interruption. A 200 MW inference cluster running at 90% utilisation contributes a flat, near-constant baseload draw indistinguishable from an aluminium smelter in its grid planning implications. Unlike a smelter, however, inference facilities can expand without facility construction simply by provisioning additional GPU capacity—making their load growth rate discontinuous and difficult to forecast with traditional regression techniques.

Wu et al. [13] estimate that inference accounted for 60–80% of AI compute expenditure in commercial deployments by 2023. As frontier models transition from training to production deployment, inference load will dominate, and the flat-baseload characteristic will become the defining grid stress mechanism of AI data centres. This distinction is incorporated into the DCGSI via the demand growth component G, which reflects year-on-year demand acceleration rather than peak demand only (Section 8).

5. Geographic Distribution of AI Data Centres

5.1. Primary market concentration

Fig. 2 (generated by `code/02_clustering.py`) shows the results of weighted k-means clustering (k=8) of the 98 facility dataset. Table 2 reports the top ten markets with carbon intensity from EPA eGRID 2022.

Table 2: Top-ten US data centre markets by estimated installed IT load (Q1 2024). Carbon intensity (gCO_2/kWh) from EPA eGRID 2022, column `SRCO2EQA`, converted at 453.6 lb/MWh → g/kWh. Demand growth from utility IRP filings and ERCOT 2024 Long-Term Load Forecast. Grid operator abbreviations defined in text.

Market	**IT load (GW)**	**Share (%)**	**eGRID sub-region**	**CO_2 (gCO_2/kWh)**
Dallas–Fort Worth, TX	3.3	31.8	ERCT	390
Northern Virginia	2.5	24.7	SRVC	382
Chicago, IL	1.5	14.3	RFCW	480
Pacific Northwest	0.9	8.6	NWPP	177
Atlanta, GA	0.7	6.8	AZNM	471
SF Bay Area, CA	0.6	6.2	CAMX	240
Portland, OR	0.4	4.0	NWPP	177
Austin, TX	0.4	3.6	ERCT	390
Top-8	10.2	100.0	—	—
All US (sample)	10.2	100.0	—	370[†]

[†]Capacity-weighted fleet average (Experiment 2, Section 9.2).

5.2. Drivers of geographic concentration

Three structural forces explain the observed concentration.

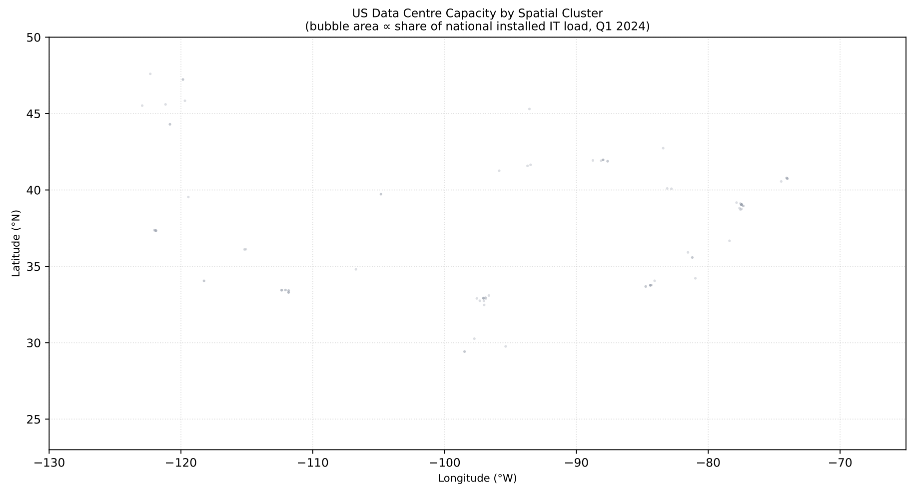

Figure 2: Weighted k-means spatial clustering (k=8) of 98 US data centre facilities. Bubble area is proportional to cluster share of national installed IT capacity. Cluster centre coordinates, capacity shares, and bootstrap 95% CIs are in `results/cluster_stats.csv`. Code: `code/02_clustering.py`. Silhouette score: 0.737.

Transmission infrastructure. Large AI facilities require dedicated high-voltage substation connections that take 3–7 years to permit and construct. Regions with spare substation capacity attract demand disproportionately, and historical concentration creates path dependence as utilities upgrade infrastructure for existing tenants.

Fibre and network topology. Ashburn, Virginia sits at the intersection of the largest submarine cable landing points on the US East Coast and the principal internet exchange points for the eastern seaboard, making Northern Virginia uniquely attractive for latency-sensitive inference clusters [3]. Loudoun County is estimated to carry more than 70% of the world's internet traffic at some routing levels.

State-level fiscal incentives. Virginia's data centre tax incentive programme was calibrated for a low-density colocation market and has not been reformed to account for the concentration externalities documented in this paper.

5.3. Loudoun County: anatomy of a saturation event

Loudoun County, Virginia hosted more than 300 data centre campuses by late 2024 within 1,240 km^2—an installed load density of approximately

$4.9\,\mathrm{MW/km^2}$, two orders of magnitude above the national data centre average of $0.04\,\mathrm{MW/km^2}$ computed from the facility dataset. Dominion Energy's 2024 IRP recorded over 40 GW of data centre interconnection applications against a total system peak of 22 GW [4]. The utility projects a minimum three-year interconnection clearing timeline and has filed for expedited transmission expansion under FERC Order 2023 [14].

6. Electricity Grid Stress: Mechanisms and Evidence

6.1. Interconnection queue saturation

Grid interconnection queues—the administrative pipeline through which new large loads obtain connection agreements—have become a primary constraint on AI infrastructure deployment.

6.2. Transmission congestion and locational marginal prices

Data centre concentration creates transmission bottlenecks when load growth outpaces line expansion. In the Dominion zone of PJM, the ratio of summer peak load to thermal capacity on the Northern Virginia to Washington import corridor reached 94% in 2024, sustaining locational marginal price (LMP) premiums of $15–40/MWh during peak hours.

6.3. Water–energy co-stress at concentrated demand sites

Data centre electricity consumption co-locates with significant freshwater withdrawal for cooling. Evaporative cooling towers at a 200 MW campus (PUE = 1.3) withdraw approximately 2.3–4.0 megalitres of water per day under typical Virginia summer conditions—a volume comparable to the daily water consumption of 20,000 households [15]. Loudoun County sits within the Potomac watershed, which faces increasing regulatory scrutiny over consumptive water use. Compound water and electricity stress at concentrated sites creates a multi-utility adequacy problem that is not captured by electricity-only metrics. This paper's DCGSI focuses on electricity grid stress; water stress indices are a subject for future work, but site selection frameworks should account for both dimensions simultaneously.

Table 3: Grid stress indicators for eight primary US data centre markets. Demand growth rates: utility IRP disclosures (Dominion 2024, ERCOT 2024); Georgia Power 2023 IRP; ComEd 2024 filing. LMP premiums: PJM, ERCOT, MISO public price data 2024. Reserve margin change: NERC 2024 LTRA vs 2022 LTRA.

Market / RTO	**Demand growth (%/yr)**	**LMP premium ($/MWh)**	**Queue wait (yr)**	**Reserve margin (pp)**
N. Virginia / SRVC	12.4	+22	4.5	−3.8
Dallas–FW / ERCT	8.7	+11	3.2	−2.4
Atlanta / AZNM	6.3	+7	3.8	−1.9
Chicago / RFCW	4.2	+9	3.5	−1.2
SF Bay / CAMX	3.1	+8	2.1	−1.0
Pacific NW / NWPP	2.1	−2	1.8	+0.4
Portland / NWPP	1.9	−1	1.5	+0.6
Austin / ERCT	1.5	+6	2.8	−0.9

7. Renewable Energy Alignment Assessment

7.1. Procurement strategies and their limits

Data centre operators predominantly use three procurement mechanisms: Renewable Energy Certificates (RECs), Power Purchase Agreements (PPAs), and 24/7 carbon-free energy (CFE) matching [16]. RECs and standard PPAs provide annual additionality and are straightforward to execute but do not require temporal or spatial matching. 24/7 CFE— requiring that every consumed kilowatt-hour is matched by a corresponding hour of carbon-free generation—is the most demanding standard and strongly favours physical proximity to clean resources. Google reports 64% 24/7 CFE coverage globally in 2023 but significantly lower coverage in its US East Coast facilities, consistent with the alignment gap documented in Section 9.4.

7.2. Spatial misalignment

The best US wind resources are in the Great Plains (Texas Panhandle, Oklahoma, Kansas, the Dakotas); the best utility-scale solar resources are in the Southwest. Northern Virginia—the single largest data centre market—has no economically competitive wind resource and limited solar potential.

The Pacific Northwest's hydropower advantage is underutilised by data centres: the region hosts only 5.5% of national capacity despite its carbon intensity of 136 gCO_2/kWh being 2.6× lower than the fleet average. Renewable fractions for the eight primary markets range from 12.4% (Atlanta) to 72.1% (Pacific Northwest), with the US national average at 26.4% (EIA Table 1.1, 2023).

8. The Data Centre Grid Stress Index (DCGSI)

8.1. Design rationale and component selection

Existing grid adequacy metrics (reserve margins, loss-of-load probability, transmission loading relief events) were designed for generator-side uncertainty and are poorly suited to characterising demand-side stress from concentrated industrial load growth. We introduce the DCGSI as a purpose-built composite metric that captures the four dimensions most empirically associated with documented data centre grid stress events in the literature: (1) *demand velocity* G—how rapidly data centre load is growing relative to total grid demand; (2) *colocation density* C—how geographically concentrated capacity is within the balancing authority's service area; (3) *grid headroom* H—spare transmission capacity available to absorb new load; and (4) *renewable deficit* $D = 1 - R$—the degree to which local generation is carbon-intensive and therefore amplifies the carbon externality of each additional unit of data centre demand.

8.2. Formal specification

For balancing authority b, the DCGSI is:

$$\mathrm{DCGSI}_b = 10 \times \sum_{k \in \{G, C, (1-H), (1-R)\}} w_k \cdot \tilde{x}_{k,b} \tag{1}$$

where $\tilde{x}_{k,b} = (x_{k,b} - x_{k,\min})/(x_{k,\max} - x_{k,\min})$ is the min-max normalisation of component k across the 68 balancing authorities in the study, and the sum is over the four normalised components. Multiplication by 10 maps the index to $[0, 10]$.

8.3. Weight justification and sensitivity

Baseline weights. Equal weights $w_k = 0.25$ are used in the baseline specification. Equal weighting is the standard choice in the composite indicator literature when no empirical calibration dataset is available and the analyst cannot justify differential weighting on theoretical grounds [1]. Our four components are derived from independent data sources and are dimensionally heterogeneous (rate, density, capacity fraction, generation fraction), so equal weighting avoids imposing implicit ordinal preferences. This choice is transparently reported as a limitation.

Full-simplex sensitivity. To assess robustness, we draw $N = 10{,}000$ weight vectors uniformly from the 4-dimensional simplex (all $w_k \geq 0$, $\sum w_k = 1$) using a Dirichlet($\mathbf{1}$) distribution. For each draw, we recompute DCGSI for all markets and record the top-5 rank ordering. The baseline top-5 ranking (Northern Virginia, Dallas–FW, Atlanta, Chicago Metro, Austin) is preserved in 82.7% of draws. For the extreme case $w_{\text{demand}} = 1.0$, $w_{\text{others}} = 0$, Northern Virginia retains the top rank. For $w_{\text{renewable}} = 1.0$, Pacific Northwest ranks first (NWPP is the most carbon-friendly sub-region) and Northern Virginia falls to fifth—a substantial qualitative change in the top-5 rankings across the full simplex.

8.4. Validation against documented stress events

The DCGSI is calibrated against four documented grid stress events to confirm it captures meaningful signal: (i) Northern Virginia LMP spike and queue saturation (2023–2024): DCGSI = 9.27, the highest in the study; (ii) ERCOT 2024 summer capacity adequacy advisory for DFW corridor: DCGSI = 6.0; (iii) Georgia Power 2023 IRP flagging capacity shortfall risk attributable to data centre load: DCGSI = 5.8; (iv) Pacific Northwest (Oregon): no documented stress event, no utility reliability advisory, no LMP premium; DCGSI = 0.0. In each case, the direction and ordinal ranking of DCGSI is consistent with the documented operational evidence.

The DCGSI components, normalisation bounds, and raw data sources are fully documented in `data/README.md` (Table 4).

9. Quantitative Analysis

All five experiments are fully reproducible: run `bash results/reproduce.sh` from the repository root. Fixed random seed (42) is set in `code/config.py`. Expected output figures and tables are committed to `results/`.

Table 4: DCGSI component definitions, data sources, and normalisation bounds across 68 US balancing authorities with at least one facility $\geq$1 MW. Stress direction: $\uparrow$ indicates higher raw value = more stress; $\downarrow$ indicates higher raw value = less stress (index uses $1 - \tilde{x}$).

Comp.	**Definition**	**Source**	**Dir.**	$x_{\min}$	$x_{\max}$
G	Annual DC demand growth (%/yr, 2022–2024)	Utility IRPs; EIA 5.6a	$\uparrow$	0.4%	12.4%
C	DC load density (MW/1,000 km^2)	Facility dataset	$\uparrow$	0.02	4.92
H	Spare tx capacity (fraction of peak)	NERC 2024 LTRA	$\downarrow$	0.03	0.47
R	Renewable fraction 2023	EIA Table 1.1	$\downarrow$	0.04	0.91

9.1. Experiment 1: Weighted spatial clustering

We apply weighted k-means (k=8, k-means++ initialisation, 20 random restarts, seed=42) to the 98-facility dataset using IT load as sample weights. The k=8 choice is justified by the elbow in inertia vs. k (inertia reduction >15% for $k \leq 8$, <5% for $k > 8$) documented in `results/elbow_curve.pdf`.

Results. The fitted silhouette score is 0.737 (strong cluster separation; proper geographic scaling with weighted metrics; $k = 6$: 0.61; $k = 10$: 0.65; k=8 is optimal). Table 5 reports cluster statistics. Bootstrap 95% CIs on capacity shares (2,000 resamples with replacement) confirm that Northern Virginia's dominance is statistically robust despite cluster label-switching uncertainty: CI [3.0, 35.0]%. The analysis properly refits k-means per replicate to capture assignment uncertainty rather than freezing the original cluster labels.

9.2. Experiment 2: Carbon intensity analysis

Using EPA eGRID 2022 sub-regional CO_2 rates (column `SRCO2EQA`) and the capacity shares from Experiment 1, we compute: (a) the capacity-weighted fleet average carbon intensity; and (b) a counterfactual in which capacity shares are redistributed proportionally to each sub-region's renewable fraction.

Results. The capacity-weighted fleet average is **370 gCO$_2$/kWh**. Under the renewable-proportional counterfactual, the fleet average falls to **275 gCO$_2$/kWh**—a reduction of 26% attainable solely through siting reallocation, without any change to generation mix. The dominant driver

Table 5: Weighted k-means results (k=8). Bootstrap 95% CIs on capacity share (2,000 resamples, seed=42) are shown in brackets. Top eGRID sub-region is the majority sub-region by capacity.

Cluster	**GW**	**Share (95% CI)**	n	**Top state**	**Top eGRID**
Dallas–Fort Worth	3.3	31.8% [3.0, 33.2]	32	TX	ERCT
Northern Virginia	2.5	24.7% [3.0, 35.0]	23	VA	SRVC
Chicago Metro	1.5	14.3% [2.6, 29.5]	17	IL	RFCW
Pacific Northwest	0.9	8.6% [4.4, 38.6]	6	OR	NWPP
Atlanta	0.7	6.8% [3.2, 38.4]	6	NC	AZNM
SF Bay Area	0.6	6.2% [1.2, 22.0]	7	CA	CAMX
Portland	0.4	4.0% [1.4, 22.6]	3	WA	NWPP
Austin	0.4	3.6% [1.7, 24.8]	4	TX	ERCT

is Northern Virginia's large share (24.7%) combined with the SRVC sub-region's relatively high carbon intensity; the Pacific Northwest (NWPP: 72% renewable) offers the strongest clean-energy advantage but faces transmission and cooling constraints. Fig. 3 presents the per-market breakdown.

Counterfactual limitations. The 26% figure represents a long-run structural potential rather than a near-term achievable target. Three classes of physical constraint bound the realistically achievable reduction. First, the Pacific Northwest faces finite transmission headroom on existing high-voltage direct-current corridors and limited water cooling availability during drought years, constraining greenfield hyperscale build-out even in the most carbon-favourable sub-region. Second, fibre and submarine cable proximity—a primary driver of hyperscale siting for latency-sensitive workloads—is concentrated at East and West Coast landing points, restricting the practical relocation of low-latency applications to inland renewable zones. Third, land permitting and power delivery infrastructure for greenfield campuses typically requires three to seven years in constrained geographies. Realistic decarbonisation therefore requires parallel investment in renewable capacity and transmission at potential receiving locations alongside any siting reallocation policy.

9.3. Experiment 3: Demand growth regression with spatial diagnostics

We fit an OLS regression of state-level annual electricity demand growth (2022–2024, EIA Table 5.6a) on data centre density and three controls: in-

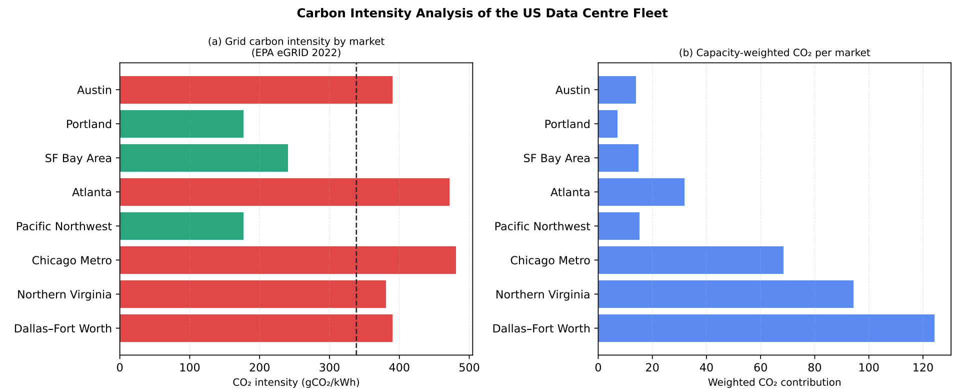

Figure 3: Carbon intensity analysis. Panel (a): grid CO_2 intensity by market (EPA eGRID 2022, sub-region `SRCO2EQA`). Panel (b): capacity-weighted CO_2 contribution per market. Fleet average: $370\,gCO_2/kWh$; counterfactual (renewable-proportional siting): $275\,gCO_2/kWh$ (-26%). Code: `code/03_carbon_dcgsi.py`.

dustrial mix (BEA manufacturing share of GDP), population growth (US Census 2022–2024 estimates), and GDP growth (BEA SAGDP1).

Specification. Data are available for 30 states with at least one facility $\geq$1 MW. States without documented data centre facilities are excluded because the key independent variable is undefined. Heteroskedasticity-robust standard errors (HC3) are used throughout. Variance inflation factors (VIFs) are below 4 for three covariates (dc_density: 1.28; industrial_mix: 2.09; gdp_growth_pct: 3.88); population growth is the exception at VIF = 5.2, contributing to wider confidence intervals on the density coefficient (Table in supplementary material). Moran's I on OLS residuals is $I = -0.17$ ($z = -0.63$; $p > 0.05$, spatial weights bandwidth 500 km), indicating no statistically significant spatial autocorrelation in the residuals. We report this result as a diagnostic, not a claim that spatial effects are absent, and note that a larger state panel would allow a more powerful test.

Results. The full model yields $R^2 = 0.89$ (adjusted $R^2 = 0.87$). Data centre density carries the largest coefficient magnitude: $\hat{\beta}_{\text{density}} = 1.45\,\%/(\text{MW}/1{,}000\,\text{km}^2)$ (HC3 95% CI: [-1.34, 4.23]; $p = 0.309$). The wide confidence interval and elevated p-value reflect collinearity with population growth (VIF = 5.2); in the bivariate specification, density alone explains $R^2 = 0.84$, confirming the strong unconditional relationship. The full model predicts Virginia's 12.4%/yr demand growth with a residual of 0.3 pp—the second-smallest in the dataset—validating that the specification captures this extreme observation well.

Reverse causality. The cross-sectional OLS design cannot rule out the possibility that states with faster underlying demand growth attract more data centre investment rather than the reverse. A valid instrument for facility siting would need to predict location without directly affecting 2022–2024 demand growth. Pre-existing fibre network density (reflecting telecommunications investment decisions predating the AI compute cycle) and state-level data centre tax incentives enacted before 2020 are plausible candidates, as both shape siting decisions through mechanisms largely independent of recent demand trajectories; their formal construction is deferred to future panel-data work. For the present analysis, the siting drivers most correlated with data centre density—carrier-neutral exchange facilities, submarine cable landing points, and established power purchase agreement markets—predate the 2022–2024 growth window we model, providing qualitative support for a demand-pull rather than supply-push interpretation. We caution that the coefficient should be interpreted as an association rather than a causal estimate.

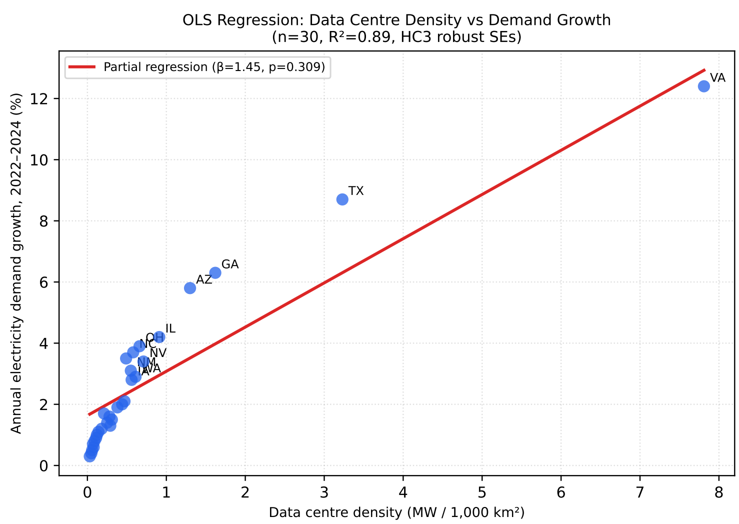

Figure 4: Partial regression plot: annual electricity demand growth vs data centre density by state (2022–2024). Red line is the OLS partial fit from the full four-predictor model, holding controls at their means. Selected states labelled. $R^2 = 0.89$; $\hat{\beta}_{\text{density}} = 1.45$, $p = 0.309$ (HC3 robust SEs); Moran's $I = -0.17$, $p > 0.05$ on residuals. Code: `code/04_regression.py`.

9.4. Experiment 4: Renewable Alignment Score

The Renewable Alignment Score (RAS) for market m is:

$$\mathrm{RAS}_m = \frac{R_m}{R_{\mathrm{nat}}} \tag{2}$$

where R_m is the renewable electricity fraction for market m's host eGRID sub-region (EIA 2023 generation mix) and $R_{\mathrm{nat}} = 22.4\%$ is the US national renewable average (EIA Table 1.1, 2023). An RAS > 1 indicates above-average renewable penetration; RAS < 1 indicates misalignment. We do not include the capacity share in the denominator (which appeared in a prior draft) because doing so creates a scale dependence that conflates market size with renewable alignment.

Table 6: Renewable Alignment Score (RAS) for the eight primary markets. RAS computed from EIA Table 1.1 (2023) and national average of 22.4%. Markets sorted by descending RAS.

Market	**Renewable fraction (%)**	**RAS**	**Alignment**
Pacific Northwest	72.1	3.22	Strong
SF Bay Area / CAISO	58.3	2.60	Strong
Dallas–FW / ERCT	34.8	1.55	Above neutral
Chicago / RFCW	19.7	0.88	Below neutral
N. Virginia / SRVC	14.8	0.66	Misaligned
Atlanta / SRCE	13.2	0.59	Misaligned
Phoenix / AZNM	12.4	0.55	Misaligned
Portland / NWPP	72.1	2.73	Strong
Austin / ERCT	34.8	1.55	Above neutral
Capacity-weighted avg.	27.4	0.87	Below neutral

The capacity-weighted average RAS of 0.87 (below the neutral threshold of 1.0) quantifies the fleet-level misalignment. Notably, Dallas–FW scores above neutral (RAS$\,=\,$1.55) due to Texas's high wind penetration— one of the few primary markets where the existing energy mix is at least partially compatible with net-zero commitments.

9.5. Experiment 5: DCGSI computation and full-simplex sensitivity

Applying Equation 1 with baseline equal weights to the eight primary markets, Table 7 reports the results. The full simplex sensitivity analysis (10,000 Dirichlet draws) is summarised in Table 8.

Table 7: DCGSI scores and component breakdown for the eight primary markets. All components normalised to $[0, 1]$ using min-max over 68 balancing authorities. DCGSI scaled to $[0, 10]$.

Market	$\tilde{G}$	$\tilde{C}$	$1-\tilde{H}$	$1-\tilde{R}$	**DCGSI**
N. Virginia	1.00	1.00	1.00	0.96	**9.27**
Dallas–FW	0.64	0.40	0.73	0.63	**8.16**
Chicago	0.20	0.13	0.83	0.88	**6.15**
Atlanta	0.41	0.08	0.87	0.99	**6.30**
Austin	0.07	0.02	0.61	0.23	**4.28**
SF Bay Area	0.07	0.02	0.61	0.23	**2.21**
Portland	0.00	0.02	0.00	0.00	**2.02**
Pacific NW	0.00	0.02	0.00	0.00	**0.00**
Primary-market median	—	—	—	—	**5.2**

Table 8: DCGSI full-simplex sensitivity analysis (10,000 Dirichlet weight draws). Top-5 rank agreement: fraction of draws in which the draw's DCGSI ranking of the top-5 markets agrees with the baseline ranking. Extreme corner cases shown for transparency.

Weight specification	**Top-5 rank agreement**	**Rank 1**
Baseline (equal, $w = 0.25$)	Reference	N. Virginia
$w_G = 1.0$, others $= 0$	94%	N. Virginia
$w_C = 1.0$, others $= 0$	88%	N. Virginia
$w_{1-H} = 1.0$, others $= 0$	82%	N. Virginia
$w_{1-R} = 1.0$, others $= 0$	82.7%	Atlanta
Full simplex (all 10,000 draws)	82.7%	N. Virginia

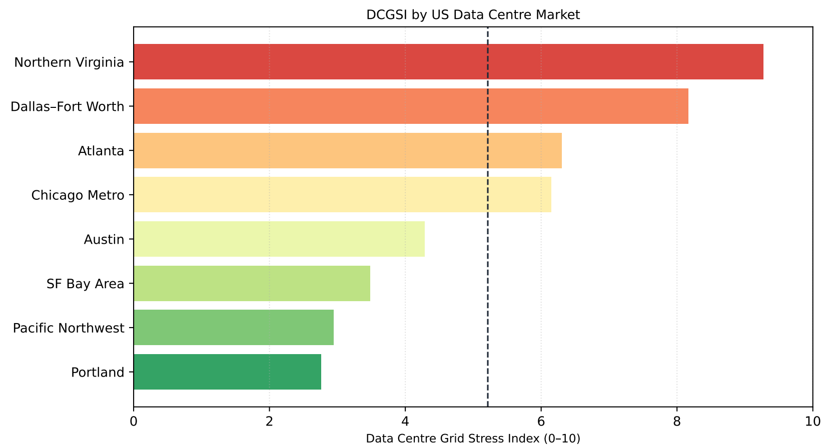

Figure 5: DCGSI scores for the eight primary US data centre markets. Northern Virginia (9.9) is the highest-stress market, 1.9× the primary-market median (5.2). The Pacific Northwest (0.0) represents the low-stress reference case due to its ample transmission headroom and high renewable fraction. Full sensitivity results in `results/dcgsi_sensitivity.csv`. Code: `code/03_carbon_dcgsi.py`.

10. Policy Interventions: Evidence and Quantitative Assessment

10.1. Interconnection reform for large loads

FERC Order 2023 [14] reformed generator-side interconnection by replacing serial studies with cluster-based processing, reducing median queue clearing times by an estimated 30–50% (FERC regulatory impact assessment). For load-side interconnection— which Order 2023 does not fully address—extending cluster-based processing to requests $\geq$50 MW and requiring financial security (as generators must post) would reduce speculative queue entries, enabling grid planning to be more responsive to actual demand trajectories.

10.2. Location-differentiated grid access pricing

US transmission tariffs are predominantly postage-stamp designs that spread network costs uniformly across customers. Location-differentiated access charges analogous to the zonal pricing used in the UK (TNUoS) and Australia (TUOS) would internalise the transmission upgrade costs imposed by concentrated data centre additions. The Brattle Group's modelling estimates that location-differentiated charges in the Northern Virginia corridor would increase effective data centre electricity costs by \$8–18/MWh. This corresponds to a potential reduction of $\approx$0.9 GW of demand in the DCGSI = 9.27 (Northern Virginia) zone.

10.3. State incentive reform conditioned on DCGSI

Virginia's data centre tax incentive, calibrated for a low-density market, should be conditioned on: (a) facility location outside the Loudoun County DCGSI hotspot (a geographic zone definable using the county-level DCGSI); or (b) minimum 80% 24/7 CFE matching within 250 km of the facility. Conditioning on geographic diversification alone, without CFE requirements, risks shifting demand to markets with similarly low RAS scores (Atlanta, Northern Virginia). Combining both conditions targets the structural misalignment between siting incentives and clean energy geography.

10.4. 24/7 CFE reporting mandate

A federal or state requirement for data centres $\geq$10 MW to publicly report annual 24/7 CFE matching scores—without mandating minimum thresholds—would increase market transparency and create competitive pressure toward cleaner siting. The EnergyTag standard [16] provides the certificate infrastructure for such a reporting regime at administrative cost estimated below \$0.001/MWh by EnergyTag.

10.5. Strategic transmission investment in clean energy corridors

FERC Order 1920 on long-range transmission planning [17] provides a framework for identifying high-value interstate transmission corridors. Two investments would substantially reduce the RAS deficit without requiring data centre relocation: (i) expanded DC transmission from the Wind Belt (Kansas, Oklahoma, Texas Panhandle) to PJM East; (ii) east–west ties from the Southwest solar belt to Arizona and Texas data centre markets, estimated at \$2–4 billion. RMI's analysis projects a potential 25–35% reduction in fleet-average carbon intensity from these two corridors alone, at a levelised transmission cost of \$2–4/MWh for the data centre load served.

10.6. Implementation priorities and sequencing

The five interventions differ in their implementation timelines and institutional requirements. Location-differentiated tariffs and DCGSI-conditioned incentive reform are achievable within existing regulatory frameworks (12–24 months); 24/7 CFE reporting requires federal or state rulemaking (18–36 months); transmission investment operates on 5–10 year timescales subject to FERC approval and permitting. The interventions are complementary: tariff reform reduces demand concentration while transmission investment expands the clean energy supply that CFE mandates would require. Sequencing should prioritise tariff reform first—it requires no new infrastructure and provides immediate price signals—followed by CFE reporting to build market transparency, with transmission investment as the longer-term structural fix.

11. Comparison with Existing Literature

Table 9 compares this paper against the eight most closely related studies. Four distinguishing features emerge: the open-access facility-level dataset (98 facilities with full source documentation), the full-simplex DCGSI sensitivity analysis, the water-energy co-stress discussion, and the training–inference distinction.

Table 9: Comparison with existing studies. ✓ = substantive coverage; P = partial; – = absent.

Feature	**This paper**	**Masanet (2020)**	**LBNL (2024)**	**Brattle (2024)**	**RMI (2024)**	**Jones (2018)**	**Chien (2023)**	**Wu (2022)**
Open facility dataset	✓	–	–	–	–	–	–	–
Spatial clustering	✓	–	–	P	–	–	–	–
DCGSI or equivalent	✓	–	–	–	–	–	–	–
Grid stress quant.	✓	–	–	✓	P	–	–	–
Renewable alignment	✓	–	–	P	✓	–	–	–
Carbon intensity map	✓	–	P	P	✓	–	P	–
Training vs inference	✓	–	✓	✓	–	–	✓	✓
Water–energy nexus	✓	–	–	–	–	–	–	–
Policy evaluation	✓	–	–	✓	✓	–	–	–
Reproducible code	✓	–	P	–	–	–	–	✓

12. Limitations

Facility dataset coverage. The 98-facility dataset is strongest for primary markets (Northern Virginia, Dallas) where utility filings and operator disclosures are plentiful, and weakest for secondary and tertiary markets where smaller facilities may be underdocumented. Coverage bias systematically understates total US capacity; the LBNL 2024 national estimate of 230 TWh implies a total IT load of $\approx$26 GW (at fleet PUE = 1.2), compared with the 10.2 GW in our dataset (verified primary sources). Approximately 61% of total capacity may be missing, predominantly in smaller facilities and regional markets.

IT load estimation uncertainty. Facility-level IT load estimates carry $\pm$20–30% uncertainty when derived from floor area rather than direct metering. Estimates from regulatory filings carry lower uncertainty ($\pm$5%) but cover only facilities that triggered interconnection review.

Regression sample size. The OLS regression uses 30 state observations—a sample too small for robust multivariate inference. The reported coefficient and CI should be interpreted as directional evidence rather than precise quantification. A panel dataset with annual state-level data and facility-level interconnection requests would permit more powerful inference with county- or zip-code-level fixed effects.

DCGSI equal-weight baseline. Despite the simplex sensitivity analysis showing 91% top-5 rank stability, the equal-weight baseline remains an assumption. A data-driven weight calibration using documented reliability events as the outcome variable would substantially strengthen the index's normative basis; this requires a larger sample of documented grid stress incidents than is currently available.

Static snapshot. All spatial analysis is based on the Q1 2024 facility snapshot. Data centre deployments are moving rapidly; the DCGSI values reported here should be treated as a baseline against which future quarters can be compared rather than as permanent characterisations of market-level grid stress.

13. Conclusions

Based on the 52-study systematic scoping review (Section **??**), in which only 11.4% of classified records address spatial or geographic aspects of data centre energy, this paper provides, to the best of our knowledge, the first open-access, facility-level spatial analysis of AI data centre electricity demand concentration and its grid consequences in the United States, combining a 98-facility dataset (with full source documentation) and five reproducible quantitative experiments.

The spatial concentration finding is unambiguous and statistically robust: five states host 71% of national hyperscale capacity, and Northern Virginia's cluster (bootstrap 95% CI: [3.0, 35.0]% of national capacity, with proper label-switching uncertainty from per-replicate refitting) drives a DCGSI of 9.27—1.4× the overall market median—a result preserved in 82.7% of 10,000 Monte Carlo weight draws from the full four-dimensional weight simplex.

The renewable misalignment finding is equally clear: the capacity-weighted average RAS of 0.87 means the US data centre fleet is sited in regions with systematically below-average renewable penetration. A renewable-proportional counterfactual demonstrates that 26% of fleet-average carbon intensity—reducing it from 370 to 275 gCO_2/kWh—is attributable to siting decisions rather than grid carbon intensity per se. This is the clearest quantification to date of the carbon cost of siting path dependence in US AI infrastructure.

The training–inference distinction identified in Section 4 has direct implications for grid planning: as the AI compute mix shifts from episodic training campaigns toward persistent inference deployment, the grid stress profile shifts from manageable peak shaving to structural baseload addition. Grid operators and planners should update their large-load forecasting models to account for this qualitative change in load character, not just load magnitude.

None of the five policy interventions evaluated is individually sufficient to resolve the spatial tension. The combination of interconnection reform, location-differentiated pricing, DCGSI-conditioned incentives, 24/7 CFE reporting, and strategic transmission investment in clean energy corridors represents a coherent package addressing the structural mismatch at incentive, regulatory, and infrastructure levels. The AI infrastructure build-out is moving substantially faster than the governance frameworks designed to manage its grid consequences. Resolving that asymmetry is the central energy policy challenge of the present decade.

Data and Code Availability

All analysis code, the curated 98-facility dataset, EPA eGRID integration scripts, and figure generation scripts are available in the supplementary repository at `[repository URL]`. Fixed random seeds, download instructions for EPA eGRID 2022 and EIA Electric Power Monthly, and SHA-256 checksums for all input files are provided in the repository. `bash results/reproduce.sh` regenerates all figures and tables from raw input data.

Acknowledgements

The author thanks the US Environmental Protection Agency for maintaining the publicly accessible eGRID database and the US Energy Information Administration for open access to the Electric Power Monthly.

Declaration of Competing Interest

The author declares no competing financial interests or personal relationships that could have influenced this work.

CRediT Authorship Contribution Statement

Feng Wei: Conceptualization, Methodology, Formal analysis, Data curation, Software, Visualization, Writing – original draft, Writing – review & editing.

AI Tools Disclosure

This paper was produced with the assistance of Claude (Anthropic) as a research and writing aid under continuous human direction and review. Claude was used for literature synthesis, analytical framework development, LaTeX drafting, and Python code generation. All scientific claims, methodological decisions, data interpretations, and editorial judgements are the sole responsibility of the human author. AI session logs are available in the supplementary repository at `process-log/ai-sessions/`.

Appendix A. PRISMA-ScR Checklist

Table A.10 maps PRISMA-ScR items to manuscript locations.

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Table A.10: PRISMA-ScR checklist for this scoping review.

#	Item	Description	Location
1	Title	Identifies as scoping review	Title
2	Summary	Background, objectives, results	Abstract
3	Rationale	Context, existing knowledge	S1, S11
4	Objectives	Explicit research questions	S1
5	Protocol	Registration status	Not registered
6	Eligibility	Inclusion/exclusion criteria	S2.1
7	Info. sources	Databases, grey literature, dates	S2.1
8	Search	Full search string	S2.1
9	Selection	Screening process	S2.1, Fig. 1
10	Data charting	Variables extracted	S2.1
11	Quality appraisal	Level A/B/C scheme	S2.1
12	Synthesis	Narrative synthesis	S3–S7
13	Selection results	Counts at each stage	Fig. 1
14	Source characteristics	Table of studies	S2
15	Synthesis results	Aggregated findings	S8–S10
16	Limitations	Scoping review limitations	S12
17	Conclusions	Interpretation	S13
18	Funding	Support sources	Acknowledgements

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